

-
5. Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

Develop a program that executes the equivalent of the shell command: `ls -l | grep ".c"` using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls.

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>
int main(){
    int fd[2];
    pipe(fd);
    pid_t p1 = fork();
    if (p1 == 0)
    {
        close(fd[0]);
        dup2(fd[1], STDOUT_FILENO);
        close(fd[1]);
        execlp("ls", "ls", "-l", NULL);
        perror("exec ls");
        exit(1);
    }
    pid_t p2 = fork();

    if (p2 == 0)
    {
        close(fd[1]);
        dup2(fd[0], STDIN_FILENO);
        close(fd[0]);
        execlp("grep", "grep", ".c", NULL);
        perror("exec grep");
        exit(1);
    }
    close(fd[0]);
    close(fd[1]);

    waitpid(p1, NULL, 0);
    waitpid(p2, NULL, 0);

    return 0;
}
```

```

    return 0;
}
venkatesh@LAPTOP-LL9RQ023:~$ gcc week5.c
venkatesh@LAPTOP-LL9RQ023:~$ ./a.out
-rw-r--r-- 1 venkatesh venkatesh 290 Jul 30 03:16 a.c
-rw-r--r-- 1 venkatesh venkatesh 87 Jul 27 08:37 addition.c
-rw-r--r-- 1 venkatesh venkatesh 87 Jul 30 03:19 addition.c.save
-rw-r--r-- 1 venkatesh venkatesh 1 Jul 30 03:24 arithmethic.c
-rw-r--r-- 1 venkatesh venkatesh 295 Jul 30 05:47 arithmethic.c.save
-rw-r--r-- 1 venkatesh venkatesh 163 Aug 3 07:03 execsc.c
-rw-r--r-- 1 venkatesh venkatesh 518 Aug 3 08:02 execv.c
-rw-r--r-- 1 venkatesh venkatesh 0 Aug 4 05:01 execworking.c
-rw-r--r-- 1 venkatesh venkatesh 265 Aug 11 12:45 execworking.c.save
-rw-r--r-- 1 venkatesh venkatesh 186 Aug 11 05:37 filename.c
-rw-r--r-- 1 venkatesh venkatesh 60 Aug 17 08:00 main.c
-rw-r--r-- 1 venkatesh venkatesh 246 Aug 25 05:49 open.c
-rw-r--r-- 1 venkatesh venkatesh 668 Aug 25 05:51 openmult.c
-rw-r--r-- 1 venkatesh venkatesh 229 Jul 28 05:42 operations.c
-rw-r--r-- 1 venkatesh venkatesh 512 Aug 11 05:34 pid.c
-rw-r--r-- 1 venkatesh venkatesh 1 Aug 4 05:23 pid.c.save
-rw-r--r-- 1 venkatesh venkatesh 149 Jul 30 06:06 process1.c
-rw-r--r-- 1 venkatesh venkatesh 190 Aug 1 09:22 process2.c
-rw-r--r-- 1 venkatesh venkatesh 1 Aug 1 09:17 process2.c.save
-rw-r--r-- 1 venkatesh venkatesh 18 Aug 1 09:18 process2.c.save2
-rw-r--r-- 1 venkatesh venkatesh 238 Aug 1 09:44 process3.c
-rw-r--r-- 1 venkatesh venkatesh 186 Aug 1 09:54 process4.c
-rw-r--r-- 1 venkatesh venkatesh 183 Aug 1 09:54 process4.c.save
-rw-r--r-- 1 venkatesh venkatesh 558 Aug 11 12:45 process5.c
-rw-r--r-- 1 venkatesh venkatesh 246 Aug 25 06:07 read.c
-rw-r--r-- 1 venkatesh venkatesh 0 Aug 17 08:08 w2.c
-rw-r--r-- 1 venkatesh venkatesh 465 Sep 3 05:13 week1.c
-rw-r--r-- 1 venkatesh venkatesh 908 Sep 3 05:24 week2.c
-rw-r--r-- 1 venkatesh venkatesh 819 Sep 3 05:25 week3.c
-rw-r--r-- 1 venkatesh venkatesh 683 Sep 3 05:31 week4.c
-rw-r--r-- 1 venkatesh venkatesh 666 Sep 3 05:38 week5.c
venkatesh@LAPTOP-LL9RQ023:~$ |

```